

EX. NO: 5

PLAYFAIR CIPHER

Aim:

To write a C program to implement the Playfair Cipher, a digraphic substitution technique that encrypts pairs of letters using a 5 x 5 key matrix.

Algorithm:

1. Start the program.
2. Construct a 5 x 5 key matrix using the keyword "PLAYFAIR", combining the letters I and J into a single cell.
3. Read a pair of plaintext letters from the user.
4. Locate the row and column positions of both letters within the key matrix.
5. If both letters lie in the same row, replace each with the letter immediately to its right, wrapping around the row if required.
6. If both letters lie in the same column, replace each with the letter immediately below it, wrapping around the column if required.
7. If the letters lie in different rows and columns, replace each letter with the one in its own row but in the column of the other letter (rectangle rule).
8. Display the resulting encrypted pair of letters.
9. Stop the program.

Program:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char matrix[5][5] = {
    {'P', 'L', 'A', 'Y', 'F'},
    {'I', 'R', 'E', 'X', 'M'},
    {'B', 'C', 'D', 'G', 'H'},
    {'K', 'N', 'O', 'Q', 'S'},
    {'T', 'U', 'V', 'W', 'Z'}
};

void find(char c, int *r, int *col) {
    if (c == 'J')
        c = 'I';
    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 5; j++) {
            if (matrix[i][j] == c) {
                *r = i;
                *col = j;
            }
        }
    }
}
```

```

    }
}
}

int main() {
    char text[100];
    printf("Enter Plain Text (2 letters): ");
    scanf("%s", text);

    int r1, c1, r2, c2;
    find(toupper(text[0]), &r1, &c1);
    find(toupper(text[1]), &r2, &c2);

    if (r1 == r2) {
        printf("Encrypted Text: %c%c\n",
            matrix[r1][(c1 + 1) % 5],
            matrix[r2][(c2 + 1) % 5]);
    } else if (c1 == c2) {
        printf("Encrypted Text: %c%c\n",
            matrix[(r1 + 1) % 5][c1],
            matrix[(r2 + 1) % 5][c2]);
    } else {
        printf("Encrypted Text: %c%c\n",
            matrix[r1][c2],
            matrix[r2][c1]);
    }
    return 0;
}

```

Output:

```

main.c
1 #include<stdio.h>
2 #include<string.h>
3 #include<ctype.h>
4
5 char matrix[5][5]={
6  {'P','L','A','Y','F'},
7  {'I','R','E','X','M'},
8  {'B','C','D','G','H'},
9  {'K','N','O','Q','S'},
10 {'T','U','V','W','Z'}
11 };
12
13 void find(char c,int *r,int *col)
14 {
15     if(c=='J')
16         c='I';
17
18     for(int i=0;i<5;i++)
19     {
20         for(int j=0;j<5;j++)
21         {

```

Output

```

Enter Plain Text (2 letters): HI
Encrypted Text: BM

=== Code Execution Successful ===

```

Result:

Thus a C program to implement the Playfair Cipher using a 5 x 5 key matrix was written, executed and the output was verified successfully.